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Modeling Conflict Using Spatial Voting Games: An Application to USDA Forest Service Lands

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Abstract

Conflict is an inherent component of natural resource management decisions in the United States and many other countries of the world. The diversity of potential uses creates a situation where individual preferences result in the need for compromise positions and coalition formation so that the natural resources can be managed. This paper demonstrates the contribution that cooperative and non-cooperative voting models can make in understanding the potential for conflict and the incentives for individuals to form coalitions. By modeling both collaborative and non-collaborative public involvement procedures, we gain insights from the differences in solutions, the implications for stability of the alternatives and the impact of the institutional power of the USDA Forest Service. The models are applied to the case of public lands management on the Shoshone National Forest.

Keywords: committee games, voting models, cooperative games, amendment agenda, the win set, the core, Shapley Value, conflict management, conflict resolution, public lands

1 Introduction

Natural resource and environmental issues are replete with decisions that generate conflict. Many believe that the conflict in the management of such resources is a result of the public "…just not understanding" the science. Others believe that the conflict is a direct result of differing values and objectives as to how the resources should be managed. Regardless of how the conflict originates, it is clear that conflict will continue to permeate natural resource decision-making, particularly in a situation where the laws and regulations require the public be involved in the decision making (for a review of the institutional issues see Martin & Bender 1999). The US Congress has mandated that agencies of the federal government consider the input of the public in any decision that may affect the environment. The objective of this paper is to model input of the public in decision-making situations regarding lands managed by the USDA Forest Service.

Although mandated to include the public in its decisions, the USDA Forest Service (USFS) can choose to undertake different types of public involvement. More recently, the USFS has embraced a collaborative (i.e., cooperative) decision-making process, which focuses on bringing stakeholders together for intensive education, communication and mediation to come to a consensus decision. In contrast, the purpose of the more traditional non-cooperative National Environmental Policy Act (NEPA) process is solely to educate interested parties about USFS activities and solicit information about the public's desires. These models focus on comparing the results of non-cooperative public involvement models with the cooperative or collaborative models.

The models address various issues associated with potential sources of conflict in managing the public lands. However, the basic source of the conflict evolves from the requirement that the diverse set of values and objectives of the various stakeholder groups must be considered in the management decision. Attempting to allocate a scarce resource, public lands, among the competing uses (i.e. commodity development, recreation, preservation, etc.) has the potential to generate a situation that appears to have no remedy. The various models presented here demonstrate compromise solutions to the conflict. The different public input procedures may have significant effects on the outcome under consideration. To understand the effect of cooperative and non-cooperative processes on the planning alternative chosen, these procedures are modeled utilizing non-cooperative voting models and cooperative committee games.

The paper is organized as follows. Section 2 presents the various cooperative and non-cooperative models. This is followed by an application of the models to a land management issue on the Shoshone National Forest in Wyoming. This section will also provide some background information that highlights sources of conflict in the management of public lands. A final section presents conclusions and policy implications.

2 The models

This section will present one non-cooperative voting models and two cooperative voting approaches for modeling both conflict management (i.e., NEPA-based, non-cooperative, public involvement) and conflict resolution (i.e., collaborative efforts). First, the non-cooperative win set will be presented, followed by the cooperative core for supramajority rule games and the cooperative Shapley Value.

In this analysis, it is assumed that the outcomes are represented in m-dimensional space and that individual preferences in this space are modeled by Euclidean distance, so that indifference curves are represented as concentric circles surrounding stakeholders' ideal points. It is further assumed that all preferences are separable. Formally, let $a \in A$ be denoted $a = (x_1, x_2, ..., x_j, ..., x_m)$, where x_j is the characteristic of a on the jth dimension. A stakeholder's preferences are separable across the m dimensions if the utility function summarizing his preferences across all alternatives in A can be expressed as $u(a) = u_1(x_1) + u_2(x_2) + ... + u_m(x_m)$. Such preferences will be used in both types of models. Each stakeholder will be referred to as i, however, on spatial representation, the analysis will denote the stakeholders' ideal points as $\{x_1,\ldots,x_n\}$ for n stakeholders. The stakeholders must choose an alternative a from a specified set $A=\{a_1,\ldots,a_n\}$ of feasible alternatives that include a default (or status quo) alternative, $y_o \in A$.

The legislative requirements, internal mandates, regulations implementing the statutes provide a confusing, though fairly discretionary policy for the USFS. The agency must consider the integrity of the ecosystem *and* manage for multiple uses. To incorporate the institutional power of the USFS within the public involvement process, the USFS is examined as either a more powerful player or as an alternative. In the first model, the win set presents the Forest Service position as a default alternative in the game. The supramajority rule game will also consider the Forest Service position as the default alternative or outcome in the analysis. The Shapley Value considers the USFS as stakeholder in the game with twice the weight of the other stakeholders.

2.1 Sophisticated voting equilibriums and the win set for non-cooperative majority rule games

The first noncooperative model considers sophisticated behavior, that is, when players have incentives to misrepresent their preferences. Shepsle and Weingast's (1984a) more formal model of the sophisticated agenda equilibrium is explained. Additionally, the win set, which builds from the sophisticated equilibria and demonstrates all the possible solutions for a backward building agenda under sophisticated voting will be explained. These techniques will consider the USFS position as an alternative that is placed in the more powerful position of being voted on last in the agenda (i.e., a backward building agenda).

The USFS has a somewhat complicated mandate to ensure that the national forest system (NFS) lands provide for multiple uses and also preserve ecosystem integrity. These mandates in themselves generate conflicting attitudes and outcomes. Additionally, many stakeholders have opinions about previous experiences and encounters with the USFS as well as other stakeholder groups. For these reasons, a model where stakeholders are sophisticated is considered. This section will focus on three aspects of sophisticated voting: the sophisticated agenda equilibrium; the win set; and the uncovered set. The uncovered set, which illustrates all those solutions possible under a forward building agenda procedure will also be shown to demonstrate the differences between forward and backward building agendas.

Shepsle and Weingast (1984a) approach the sophisticated agenda solution with a formal and more intuitive structure. It is necessary at this point to introduce some additional notation and proofs as described by Ordeshook (1986) and Shepsle and Weingast (1984a). We assume that n stakeholders, $N = \{1, \ldots, n)$, must select a single element from a convex policy space $X \subset R^{m}$ (the m dimension real space), where n is odd. The preferences of each stakeholder are given by a weak ordering $R_i(x) \subset X \times X$. Thus, $R_i(x) = \{y \in X \mid yR_ix\}$. For each $i \subset \mathrm{N}$, a distinguished point $x_i \in \mathrm{X}$, called i's ideal point, with the property that x_i is preferred to y for all y not equal to x_i ($x_i\mathrm{P}y$ $\forall\ y \neq x_i$.). $\mathrm{P}_i(x)$ and $\mathrm{I}_i(x)$ are the interior and boundary, respectively of $R_i(x)$. The former is i's preferred-to-x set and the latter is his indifferent-to-x set. Additionally, the set of alternatives $A \in R^m$ so that policy space consists of both alternatives $A = \{a_1, \ldots, a_s\}$ and stakeholder ideal points $X = \{x_1, \ldots, x_n\}$. The status quo or default alternative, y_o is also an element of R^m. Throughout the analysis, the following characteristics common to multidimensional voting models are assumed: continuity; strict quasi-concavity; and compact preferences. Additionally, let $W(x)$, defined as the win set, be the set of points the majority preferred to x: $W(x) = \{y \mid y\mathrm{P}x\}$.

Shepsle and Weingast (1984a; 1984b) have suggested and a number of others have demonstrated experimentally that procedural rules do constrain political outcomes; specifically, that because the default outcome is always considered in the final round, a backward voting procedure places a strong constraint on the collective choice. Shepsle and Weingast (1984a) have shown that for any alternative to be the final outcome, it must be the case that it is either the default outcome, y_o, (considered last) or that the alternative must be an element of the win set of y_o, that is $a_1 \in W(y_o)$.

Similarly, Shepsle and Weingast (1984a) have also demonstrated that for a set agenda in which the status quo is always considered first (i.e., forward building agenda), the set of possible outcomes are those not covered by y_o, the uncovered set, $UC(y_o)$. An alternative, a_1 is covered by y_o (y_oCa_1) if y_o P a_1 and if every alternative that is majority preferred to y_o also is majority preferred to a_1.

To illustrate the win set and the uncovered set graphically, consider figure 1. The three petals, denoted by A, B and C, comprise $W(y_o)$. For example, consider region A. This is the area that stakeholders 2 and 3 (the majority) prefer to y_o. The larger shape that's borders are outlined is $UC(y_o)$. The uncovered set by definition contains the win set, $W(y_o)$. The Pareto optimal set is the triangle connecting the three ideal points. Here, an agenda setter who must commence at point y_o may construct an agenda with any point in the UC(y_o) as the sophisticated result. An amendment agenda which must finish with y_o is constrained to be either y_o or an element of $W(y_o)$.

INSERT Figure 1: The win set and the uncovered set

2.2 The core for supramajority rule games

The second model will consider a cooperative supramajority (i.e., four-fifths) rule game where the number of players needed to decide on an alternative is greater than the majority. This type of game is relevant in situations where the USFS brings together all concerned stakeholders in a series of meetings hoping to come to a consensus decision. The greater the number of people who agree, the more likely the USFS will receive support for its actions and will not be sued by disgruntled stakeholders. Another advantage to this type of model is that there are weaker restrictions for the existence of a core solution under supramajority rule than under majority rule game.

With majority rule games, a core exists only if there is a Condorcet winner in all directions. A Condorcet winner is defined as follows. If O is the set of outcomes, then $c \in O$ is a Condoret winner if, for all other $d \in O$, the number of people who strictly prefer c to d exceeds the number who strictly prefer d to c. Supramajorities have the effect of restricting the set of winning coalitions while simultaneously admitting blocking coalitions (Laing et al. 1983). These majorities make it more difficult to upset the default alternative. Greenberg's (1979) first theorem describes the conditions necessary for a nonempty core. If we let m be the number of issues and z represent the percentage of votes required to reject the status quo and pass a specific outcome. Then formally, Greenberg (1979) demonstrates that assuming a simple n-stakeholder game with *z*-majority rule, the core is *nonempty* for every set of preferences over the alternatives if and only if:

1. $z > m/m+1$ if all outcomes are a closed, convex, and bounded subset of *m*-dimensional Euclidean space, and if preferences are convex; *or*
2. $z > (s\text{-}1)/s$ if all outcomes are a finite number of *s* alternatives.

If we assume a four/fifths majority rule, in two-dimensional issue space, since 4/5 is greater than 2/3, under Greenberg's (1979) first theorem, this game will contain a nonempty core. The quota, *q*, of this game would be 4, meaning that at least 4 players must agree to pass an alternative, otherwise the status quo prevails. This game assumes that each player's preferences are issue oriented, and that each players preference ordering of the alternatives is complete, transitive, continuous, and at least weakly convex.

Generally, the characteristic (or coalition form) is denoted by (v,N,U), where *v* is a characteristic function defined over subset *C* of *N*, the set of players, and U is the set of feasible utility outcomes. In this model, it is unnecessary to utilize the characteristic form; the utility is determined as distance (i.e., indifference curves) from the ideal point. The stakeholders must choose one outcome *o* from a specified set $O=o_1\ldots,o_v$, of feasible outcomes that include a default (or status quo) alternative, $y_o \in O$. A minimum winning coalition, W*, is a coalition in which the defection of any member, *i*, renders the remaining members $C\text{-}\{i\}$, losing or blocking. A winning coalition (W) can enact any feasible alternative, a losing coalition (L), nothing and a blocking coalition (B) can enact only the status quo. In the above example, a coalition is winning if it can secure *q* stakeholders. Thus, any coalition C is fully effective, in that it can enact any alternative *a* in its effective set $E_C \subseteq O$, where $E_C = O$ if $C \in W$, $E_C = \emptyset$ if $C \in L$, and $E_C = y_o$ if $C \in B$.

It is assumed that each stakeholder has a weak preference relation, R_i, on the set of *O* outcomes, where R_i is reflexive, complete, transitive, continuous and weakly convex. Let P_i and I_i denote strict preference and indifference, respectively. For any nonempty subset $C \subseteq N$ of stakeholders and any reference point $o \in O$, define the collective preference sets (Laing and Slotznick 1987) for weak preference,

$$R_C^O = \{o' \in O \big| (\forall i \in C) o' R_i o\},$$

and for strict preference,

$$P_C^O = \{o' \in O | (\forall i \in C) o' P_i o\}.$$

Since preferences are demonstrated with Euclidean distance, the Pareto optimal set for a coalition is the convex hull of the coalition members' ideal points. The Pareto optimal set of alternatives for any coalition may be defined as $B_C = \{o \in o | P_C^o = \varnothing\}$ or the smallest convex set of outcomes that includes the ideal points of all members of C. The feasible outcome *o* is defined to be *rational* for C if there is no feasible outcome that dominates *o* via C (that is, $P_C^o \cap E_S = \varnothing$). The heartland H is defined to be that subset of feasible alternatives that are rational for every winning coalition:

$$H \equiv \{o \in O | (\forall C \in W) P_C^o \cap E_C = \varnothing\} = \cap_{C \in W^*} B_C ,$$

That is, an outcome, *o*, is rational for a set C of players if C has nothing in its effective set that every member of C strictly prefers to *o*. The heartland also equals the intersection of Pareto optimal sets for the set of minimum winning coalitions, W*, where C $\in$W* if C, but no proper subset of C, is a winning coalition. The heartland is shown as *H* in Figure 2.

INSERT Figure 2: The core of a simple game with four-fifths rule

The subset of feasible alternatives is blocker rational that are not dominated by the default outcome y_o via any blocking coalition:

$$\{o \in O | (\forall C \in B) P_C^o \cap E_C = \varnothing\} = \{o \in O | (\forall C \in B) y \notin P_C^o\}.$$

The core is the set of all outcomes that are rational for every winning and blocking coalition in the game. The core, denoted *K*, is the set of all feasible outcomes that are undominated. The core in the simple collective decision game is

$$K = \{o \in O | \forall C \subseteq N) P_C^o \cap E_C = \varnothing\}.$$

The core must be both rational for every winning coalition, i.e., $K \subseteq H$, and must also be contained in the set of outcomes that are viable against the default outcome or status quo, $V(y_o)$:

$$V(y_o) = \cup_{C \in W} R_C^{y_o} = \cup_{C \in W^*} R_C^{y_o} .$$

$V(y_o)$ is the intersection of the blocking stakeholders' indifference curves (see Figure 2). Thus, the core is that subset of the heartland that is viable against the default outcome (i.e., the intersection of the heartland and the blocking stakeholders' indifference curves preferred to y_o).

In any strong simple game, the heartland is viable against the default outcome, that is, the heartland is the core. In simple committee games, if the heartland and $V(y_o)$ have no point in common, the core is empty.

2.3 Shapley Value

The third model to be examined is the Shapley Value. This type of model is used for cooperative games with transferable utility. The previous models considered non-transferable utility games. This model is investigated to attempt to emulate some of the transferable characteristics of tradeoffs concerning public land planning as well as for comparison to the nontransferable case. For example, an environmental group may not oppose an oil and gas lease on public lands if the USFS is willing to improve watersheds elsewhere. Another example is when President Clinton settled the issue of developing the New World Mine on the Wyoming and Montana border. In essence, the New World developers joined the coalition advocating "no development" in return for a considerable some of money. This scenario can be considered one of transferable utility.

It is necessary to reintroduce the characteristic function. The characteristic function is superadditive, that is, when any two coalitions unite that have no members in common, the new coalition can ensure anything for its members that the two separate coalitions could ensure separately, and perhaps more. Characteristic function (*v*, *N*, *U*) is defined on the set of all subsets of coalitions, C, of the all stakeholders, *N*, and the set of all feasible utility outcomes, *U*.

This model further assumes Euclidean preferences consistent with the other models. Therefore, utility of an associated alternative for each stakeholder group is dependent on the distance from each stakeholder's ideal point. The coalition worth, $v(C)$, will be a function of the utilities of the players. The coalition function can be calculated with the following equations (Shields et al. 1997). If C is a winning coalition (i.e., it contains a majority of the votes), then it chooses any alternative from A and can secure itself joint utility v(C), where

$$v(C) = \max_{a_j \in A} [\sum_{i \in C)} u_i(a_j)].$$

If C is a losing coalition, it can be sure of obtaining not more than v(C), where

$$v(C) = \min_{a_j \in A} [\sum_{i \in C} u_i(a_j)].$$

Further, it is assumed that the empty coalition has zero value.

The Shapley Value predicts a unique payoff allocation for all the players. Shapley (1953) approached this problem axiomatically. That is, he asked what kinds of properties we might expect such a solution concept to satisfy, and he characterized the mappings ϕ that satisfy these properties. Let $\phi(v)=(\phi_1(v),\ldots, \phi_n(v))$ denote the Shapley Value (SV) of the game defined by the characteristic function (coalition worth) v, where $\phi_i(v)$, $i=1,\ldots,n$, is the Shapley Value payoff to player i. The Shapley Value can be defined by the following three conditions (Myerson 1991).

1. *Carrier*. If C is a *carrier* of *v*, then all players who are not in C are called *dummies* in *v*, because their entry into any coalition cannot change its worth. This axiom asserts that the players in a carrier set should divide their joint worth among themselves, allocating nothing to the dummies. A coalition C is a carrier of a coalitional game, *v*, if and only if

$$v(C \cap C') = v(C) \qquad \forall C \subseteq N$$

2. *Symmetry*. In a symmetric game all players receive equal payoffs. The symmetry axiom asserts that only the role of a player in the game should matter, not his or her specific names or label in the set *N*.
3. *Linearity or Additivity*. This axiom asserts that the expected payoff to each player should not depend on whether the players bargain about coalitional strategies today (before the resolution of uncertainty) or tomorrow (after the resolution of uncertainty). For any two coalitional games, *v* and *w*, any number p such that $0 \le p \le 1$, and any player *i* in N,

$$\phi_i(pv + (1-p)w) = p\phi_i(v) + (1-p)\phi_i(w)$$

There is one function, the Shapley Value, that satisfies these axioms:

$$\phi_i(v) = \sum_{C \subset n-i} [v(C \cup \{i\}) - v(C)] \frac{|C|!(n-|C|-1)!}{n!}$$

where $|C|$ refers to the number of members in coalition C and n is the number of total stakeholders in the game.

The worth of the coalitions will depend on the utility values associated with each alternative as well as the number of people in the coalition. This value does not take into account any majority rule; the coalition values drive their formations. However, from the coalition worth functions, majority coalitions are worth more than minority coalitions. The SV will define levels of utility due to individual stakeholders. If there is an alternative that generates exactly these utility levels, then a solution has been found. If one does not exist, then the solutions that yield similar utilities to the SV will be identified as the solution.

3 An application: the Shoshone National Forest

Consider a situation where the Forest Service is evaluating a proposal for oil and gas leasing outside of Dubois, Wyoming on the Shoshone National Forest. Specifically oil and gas leases are being considered in the Brooks Lake/Lava Mountain area. Dubois is located in the Wind River Valley, in northwestern Wyoming, an area well-known for its populations of game and watchable wildlife. The Shoshone National Forest surrounds the valley on three sides. The Dubois economy has historically been driven by recreation expenditures and commodity production.

The oil and gas leasing proposal has triggered the need for an analysis of the environmental impacts (the NEPA process), which initiated considerable public involvement. The conflict surrounding this oil and gas development has been substantial. Many residents contend that local gains in terms of energy based jobs and income, and increased tax collections, would not compensate for decreases in recreation based income or amenity and life style losses. The oil industry position has been that energy resource development is a legitimate use of public lands and the area has high potential for an economic discovery. A complete discussion of the issues and economy of the area is available in Shields (1993).

Martin et al. (1997) describe seven players in the Shoshone case study. This analysis will focus on five of the seven players. The Forest Service, which is considered a player in Martin et al. (1997) will be examined as either a default alternative or a powerful stakeholder in this analysis. Also, the local tourist industry and the local governmental units are combined to represent one stakeholder bliss point in issue space, since their ordinal ranking of policy options is similar.

Based upon the available documentation five stakeholders are identified:

- x_1: Oil Company that Proposed Leasing
- x_2: Environmental Organizations
- x_3: Local Tourist Industry and Local Governmental Units
- x_4: Local Timber Industry
- x_5: Local Retail and Wholesale Merchants

This scenario encompasses many issues which are important to the conflict surrounding the proposal for oil and gas leasing on the Shoshone National Forest. This analysis will focus on two issues: 1) oil and gas development; and 2) managing for recreation in the area (measured in recreational visitor days (RVD)). The relevant bliss points of the stakeholders considering these two issues are assessed given the considerable documentation of the situation. The stakeholders' ideal points in issue space are portrayed in Figure 3. The alternatives will be discussed in the following section.

INSERT Figure 3: Stakeholders' Ideal Points

The oil and gas company, x_1, obviously prefers to drill many wells and would also like the Forest Service to focus on relatively less management for recreation, since this activity could potentially conflict with additional oil and gas development. Similar reasoning leads to the timber industry's position, x_4, of managing for less recreation and less oil and gas development since these activities could also lead to fewer acres being available for timber harvesting in the future. The environmental organizations, x_2, prefer that no drilling take place on the Shoshone National Forest whatsoever. Additionally, they want to decrease the number of RVD on the Shoshone; if there is less infrastructure to accommodate these recreationists, then fewer people will visit the forest thereby sustaining the resource.

The local tourist industry, x_3, relies on the income generated by visitors to the area, and would like to have the Forest Service manage for additional recreation. They also feel that too much development of the Brooks Lake/Lava Mountain area would hinder the attractiveness of the area and therefore would diminish the number of visitors to Dubois. The local government, also x_3, prefers some oil and gas leasing. However, because the majority of tax collections and benefits of the oil and gas development accrue to the state and nation and not to the local area, the local governments would prefer not to completely support this development. Instead the local government would prefer that the Forest Service focus on managing for more RVDs. Because these are very similar positions, they are combined into one.

The retail and wholesale industries, x_5, favor any activity which will expand the economy of Dubois. They prefer that the Forest Service allow the oil and gas development to continue but not allow all the development which may begin to hinder the tourism industry. Additionally, the retail and wholesale industries want the Forest Service to manage for quite a bit more recreation, which will encourage more visitors to the area.

The Forest Service's alternative, y_o, is dictated by the many public land laws and mandates required in the statutes and regulations. Specifically, the Mineral Policy Act of 1920, the relatively new guidelines of Ecosystem Management, and President's Bush's new energy policies all play a role in how the agency approaches these issues. The Forest Service would like to provide as many multiple use resources to the public as possible within the constraints of the ecosystem. Therefore, the agency prefers to allow the oil and gas leasing and would also like to provide more infrastructure for recreational uses of the lands.

This scenario encompasses many potential issues important to conflict surrounding leasable development on the SNF. This analysis will focus on two issues (see figure 3):

1) the percentage of forest legally available (1,003,320 acres) for oil and gas exploration and development that does not include the stipulation of "no surface occupancy" (NSO) (this issue will range from 0 to 45 percent of the acres available for analysis on the SNF); and
2) managing for both dispersed and developed recreation in the area (this issue will range from 1.6 million to 3.0 RVD).

Currently, the SNF is managed for 2.6 million RVD for dispersed and developed recreation. Additionally, the final EIS proposes that 19 percent of legally available lands without NSO stipulation be available for oil and gas exploration and development. This point is y_o in figure 3. To carry out the subsequent utility analysis on the scenario described above, it is necessary to adjust the scales on the axes so that they are compatible. Utility comparisons cannot be made unless the units of both of the "goods" are the same. For these reasons, the ranges for the scales will be normalized from 1 to 100.

Table 1: Normalized Values for Utility Analysis

	Leasable Development		Recreation	
Stakeholder Ideal Points	Percentage of Lands Available without NSO	Rescaled Values for Acres Available	Millions of RVDs	Rescaled Values for Recreation
x_1	40	89	2.2	43
x_2	28	62	2.2	43
x_3	0	0	2.3	50
x_4	10	22	2.7	78
x_5	22	49	2.9	93
y_o	19	42	2.4	57

These figures will be the ones that are discussed and analyzed (figure 3) for the remainder of this chapter.

3.1 Sophisticated voting models and the win set

The models considered in this section assume that the stakeholders have knowledge of the game, the agenda and other stakeholders' preferences (i.e., they are sophisticated). This is assumed throughout the analysis. To determine the sophisticated solutions, three types of solutions are considered in this section: the sophisticated agenda equilibrium, the win set, and a graphical analysis for the uncovered set.

Since the default alternative of the USFS position is always considered last in the voting agenda, this type of procedure is termed a backward building agenda. To illustrate the possible solutions with a backward building amendment agenda process, both a numerical and graphical analysis is demonstrated. To determine the sophisticated agenda and if it lies in the win set, the following analysis can be applied to an amendment agenda procedure (Shepsle and Weingast 1984a). Let $Z = (z_1, \ldots, z_n)$ of non-innocuous motions. This analysis involves identifying a majority preference relation table for determining the win set (see Table 2). An entry of "1" in a cell means the row alternative is preferred by a majority to the column alternative; "0" implies the opposite.

Table 2: Majority Preference Relation

	a_1	a_2	a_3	y_o
a_1	-	1	0	0
a_2	0	-	0	1
a_3	1	1	-	0
y_o	1	0	1	-

Let $V = \{a_1, a_2, a_3, y_o\}$ identify the order of the agenda under consideration, a_1 being first, a_2 second, a_3 third and y_o lastly (see Figure 3 for alternatives). The sophisticated agenda equilibrium, Z, is equal to $\{z_1, z_2, z_3, z_{yo}\}$. In this application, the sophisticated equilibrium set is determined by the subsequent analysis, which considers the last alternative first.

$z_{yo} = y_o$ since y_o is trivially noninnocuous;

$z_3 = y_o$ since $a_3 \notin W(y_o)$;

$z_2 = a_2$ since $a_2 \in W(y_o)$;

$z_1 = a_2$ since $a_1 \not\in W(a_2) \cap W(y_o)$.

The win set analysis is read through the majority preference relation. For example, to determine if a_3 is an element of y_o's win set, consider the upper right hand corner of the matrix in Table 2. Reading from a_3's row, since there is not a "1" depicting the win set in row a_3, column y_o, a_3 is not an element of y_o's win set. The sophisticated agenda is equal to $Z = \{a_2, y_o\}$. Through Theorem 1 (Shepsle and Weingast 1984a), the first element of Z, in this case, a_2 (22,71) is the *sophisticated agenda equilibrium*, which according the Shepsle and Weingast (1984a) must lie within the win set of this game. The outcome a_2 confirms our initial subgame perfect equilibrium

The following graphical analysis demonstrates the win set or all of the majority preferred sets to the default alternative y_o (Figure 4).

INSERT Figure 4: All win sets with a default alternative, y_o

There are four stakeholder majority groupings that contain points in a win set: {1,2,3} {5,1,2}, {4,5,1} and {3,4,5}. The upper left hand petal, A, in Figure 4 is associated with the preferences of the majority 3, 4, and 5; these are the points this majority prefers to y_o. Specifically, this petal is bound by stakeholder 3 and 5's indifference curves that contain the status quo. Therefore, the lower bound is y_o (42.0,57.0) and the upper bound is (12.4,90.7) with utility value of –42.6 for stakeholder 3 and a utility value of -36.7 for stakeholder 5. The larger right hand petal, B, is associated with the preferences of stakeholders 1, 2, and 5. Region B is bounded by stakeholders 2 and 5's indifference curves, specifically, y_o (42.0,57.0) on the left and (72.6,65.0) on the right. Stakeholder 2's utility is -24.4 along the upper bound and stakeholder 5's utility is –36.7 along the lower bound of this petal. The top smaller petal is associated with stakeholders 4, 5 and 1 and is bounded by stakeholders 1 and 4's indifference curves. Region C is bounded by (42.0,57.0) and (50.7,73.6) with a utility of –49.0 for stakeholder 1 and –29.0 for stakeholder 4. The bottom region, D, which represents the majority 1, 2, and 3, is bounded by (42.0,57.0) and (40.4,36.5) with a utility of –49.0 for stakeholder 1 and –42.8 for stakeholder 3.

Since alternative a_2 lies within the win set, under any kind of backward building sophisticated agenda, the solution will either be the status quo or an element of the win set (e.g., alternative a_2) (Shepsle and Weingast 1984a). With a forward building agenda (i.e., where the default alternative is considered first in the agenda), the uncovered set would contain all the possible solutions generated by all agendas starting with the default alternative, y_o. This area is much larger that that of the win set. By definition, it encompasses all the points in the win set. With the fairly centered USFS default position, the uncovered set will most likely also include all those points in the Pareto optimal set (i.e., the convex set connecting all the stakeholders' ideal points). Determining the uncovered set for 5 stakeholders is an encumbering process that will not produce insightful solutions. Further, the win set and the backward building agenda provide an institutional analysis that gives the USFS the ultimate position of being considered last in the agenda by the stakeholders.

3.2 The Supramajority Rule Game

Consider a supramajority (four-fifths) rule game with five stakeholders on the Shoshone NF. The USFS position, y_o, will be analyzed as the default or status quo position. This type of game again assumes that the blocking coalitions if effective can enact the default alternative, y_o, and losing coalitions cannot deliver any alternative. This gives the default outcome power over the other alternatives in the analysis. The heartland for this five-person game, the area undominated by any winning coalition, is the area enclosed by the dark line in Figure 5. This is the core if there are only winning and losing coalitions. However, in this scenario, blocking players can enact the default outcome, y_o.

All the stakeholders in this game have the potential to be blocking players. That is, all stakeholders do not prefer either the default outcome or the heartland. The stakeholders will only form a coalition if the solution is preferred to the default alternative, y_o. Therefore, an analysis of each stakeholder's rational region, the region that is preferred to the default outcome, must be undertaken. The only coalition that has rational areas that intersect is coalition $\{1,2,4,5\}$. All the other coalitions that have the potential to form will not because at least one stakeholder per coalition will prefer to enact the default outcome. Stakeholders 2 and 4 and their corresponding rational areas are binding the solutions within the heartland. The intersection of the heartland and the rational area for the blocking players (stakeholders 1,2,4,5) is the core for this game. Only the binding player's indifference curves are portrayed since these are the bounds of the core.

INSERT Figure 4: The core for a four-fifths supramajority rule game with a default alternative

The core is bound on the upper right hand side by the intersection of stakeholders' 2 and 4 indifference curves that contain the status quo at point (45.5,61.0). The core is constrained at the lower edge by the heartland's line between stakeholder 2 and stakeholder 4. The point at which this line crosses stakeholder 2's indifference curve is (43.6,59.1) and the point where this line crosses stakeholder 4's indifference curve is (43.8,58.9). The core then falls for acres available for oil and gas development between 43 and 46 and for recreation between 59 and 61.

3.3 The Shapley Value

The Shapley Value is examined as a game that considers the FS as a player with twice the weight of each of the other players (seven stakeholders total). Therefore, the number of coalitions that can form is 2^7, or 128. It is assumed that winning coalitions must contain *at least* a majority of stakeholders. However, the SV does not make coalition prediction; it produces utility values for each stakeholder that satisfy the axioms, carrier, symmetry and linearity.

This game will consider the four alternatives depicted in Figure 3. The characteristic function can be calculated with the following equations (Shields et al. 1997). If C is a winning coalition (i.e., it contains a majority of the votes), then it chooses any alternative from A and can secure itself joint utility v(C), where

$$v(C) = \max_{a_j \in A}[\sum_{i \in C)} u_i(a_j)].$$

If C is a losing coalition, it can be sure of obtaining not more than v(C), where

$$v(C) = \min_{a_j \in A}[\sum_{i \in C} u_i(a_j)].$$

Further, it is assumed that the empty coalition has zero value. Using Table 2 and the above equations, it is possible to determine the characteristic function of each coalition. For example, consider coalition {1,2,3}. The characteristic function is determined by the following

$$v(1,2,3) = \max_{a_j \in A}[-136.1, -132.0, -106.7, -116.0] = -106.0$$

Through iterations on a computer program (Bender 1998), the SV for the i^{th} stakeholder (i = 1,…,7) is determined to be:

$$\phi_i(v) = (-43,8,-22.9,-28.1,-15.5,-26.3,-2.0)$$

These utility levels do not relate exactly to the utility levels of one of the alternatives. Since these utility levels represent distances from the stakeholders ideal points, the intersection of all these indifference curves represents a solution. However, not all of the indifference curves intersect (Figure 6).

INSERT Figure 6: The Shapley Value for the seven stakholder game

The point that most likely bounds this section is the solution of majoirty coalition {2,3,y_o}. Since y_o has twice the weight of stakeholders 2 and 3, this solution is probably located just southwest of y_o's indifference curve. It is assumed then that this most southern bound of y_o's indifference curve (42.5,55.0) creates the line (m″=55.0) from intersection (2y_o) to the intersection of (m″=55.0) and stakholder 3's indifference curve. This solution is approximated by finding the points of intersection of the indifference curves and connecting the points. This approximate solution is the region presented below in Figure 7. These points of intersection are given in Table 3.

Table 3: The bounds of the Shapley Value solutions for the 7 stakeholder game

Intersection of stakeholder indifference curves					
12	34	45	51	2y_o	3 (m″=55)
49.7,62.3	25.0,62.8	35.5,70.4	52.3,66.9	42.5,55.0	27.7,55

INSERT Figure 7: An approximate solution for the seven stakholder Shapley Value

It is evident that as the USFS player becomes more powerful, the SV solution centers on the USFS position.

5 Solution analysis and policy implications

This paper has covered a considerable number of solution techniques to investigate the cooperative and non-cooperative planning proccesses surrounding leasable development and recreation decisions on the Shoshone National Forest in Northwest Wyoming. Each solution concept has been examined with special institutional considerations for the USFS position, which is the preferred alternative noted in the EIS and Forest Plan. This section will look at the differences within the non-cooperative models and the cooperative models and examine some of the policy implications that emerge from this analysis (see Table 4).

Table 4: Solutions for the non-cooperative and cooperative models

Solutions	Leasable Development	Recreation	**Institutional considerations**
Non-Cooperative Voting Models:			
Win set: Solution A	12-42	57-91	BB agenda, USFS as final alternative considered
Solution B	42-73	57-65	BB agenda, USFS as final alternative considered
Solution C	42-51	57-74	BB agenda, USFS as final alternative considered
Solution D	40-42	36-57	BB agenda, USFS as final alternative considered
Cooperative Games:			
Core for $4/5^{th}$ rule	43-46	59-60	USFS position considered the default alternative
SV w/USFS	25-52	55-71	USFS a player with twice the weight—alternatives considered

The non-cooperative voting model utilizes the win set to analyze the traditional NEPA process of including stakeholders in public land planning, specifically planning for leasable development and recreation. Considering sophisticated behavior and a backward-building amendment agenda, the solution is alternative a_2, (22,71). This occurs since the majority of stakeholders prefer this alternative to the USFS position. However, if an alternative lies within the win set, it has the possibility of beating the USFS alternative. Since the cooperative solution techniques do not consider an amendment agenda, it is the win set solutions that are more easily compared to the cooperative games and give perhaps more robust results.

The petals of the win set have important implications for the alternatives that the USFS considers in its EIS process. The USFS could utilize the win set as a starting point for choosing alternatives in its planning process. However, the agency could also choose alternatives not within the winset, thereby giving its "default alternative" a greater chance of becoming the final outcome. Additionally, the solutions represented by the petals of the win set are substantial, disparate, and do not fall in one region. For these reasons, the NEPA process could bring about public land decisions that are extremely dissimilar from the USFS position, which could imply solutions that are not stable policy outcomes. This non-cooperative model could therefore predict the potential for future litigation.

The cooperative games were examined to represent the collaborative decision-making process that has become a prevalent approach for the USFS in recent years. The USFS brings many of the interested stakeholders together to decide on an alternative. The cooperative games comprise a supramajority rule model for the core and the Shapley Value, which lets the (transferable) utility values drive the coalition formation.

The core for supramajority rule games relies on the fact that the blocking players can enforce the default alternative, which is examined as the USFS position. The supramajority (i.e., 4/5$^{\text{ths}}$ rule) game is convenient for modeling USFS "resolution" management decision-making as in reality, emphasis is often placed on maximizing the number of stakeholders who can agree on any given alternative. The supramajority rule game predicts that only one coalition will form {1,2,4,5} given the 4/5$^{\text{th}}$ rule game. Therefore, these solutions are more apt to be "centered" with respect to these four stakeholders. The area that they agree on is from 43-46 for leasable development and 59-60 for recreation and is quite close to the USFS default alternative (42,57). This model could help in explaining why the USFS position was the outcome of the Shoshone NF planning effort. Although a "cooperative" game, stakeholder 3, representing environmental organizations, is not part of the coalition, which could also indicate potential instability and possibly litigation.

The Shapley Value was the second cooperative model considered. Although the SV makes no coalition predictions, it is assumed that majority coalitions are worth more than minority coalitions. The SV furnishes unique (transferable) utility levels due each stakeholder. The utilities are plotted as distances from the ideal point of each stakeholder. The SV solution would be the point at which all these indifference curves intersect, however, not all of the curves intersect. (The intersection points of the indifference curves that fall within the convex set of stakeholder ideal points are noted.) These points are connected to render a region that demonstrates the approximate solutions from the game. The USFS position is given twice the weight of the other stakeholders, and the size of this solution region centers around the USFS position (25-52 for leasable development and 55-71 for recreation).

The SV, because of its condition of symmetric utilities, determines a region of solutions that are considered "fair" outcomes for all stakeholders. This solution could be a predictive solution, in that this "fair" region could be the starting point for alternative selection in the EIS process, leading then to one alternative within this region as the solution. Additionally, this region obviously contains the USFS position, which could help in explaining why this alternative was chosen by the Shoshone NF planning process.

The solutions to the models presented above demonstrate that multiple equilibria exist and that regardless of the alternative chosen, at least one player will not be satisfied. Given that at least one player or stakeholder group will not be satisfied, the possibility still exists that the USFS decision will be challenged in court. However, the likelihood of success of such a challenge would be significantly diminished given the opportunity to participate in the cooperative "game". Given the more equitable solutions of the cooperative game (i.e., the core as well as the solutions in the SV region), it is possible that this process does offer a more stable outcome. Future research could investigate, given similar outcomes under both processes, the transaction costs of the cooperative process balancing the reduced (?) potential for future litigation.

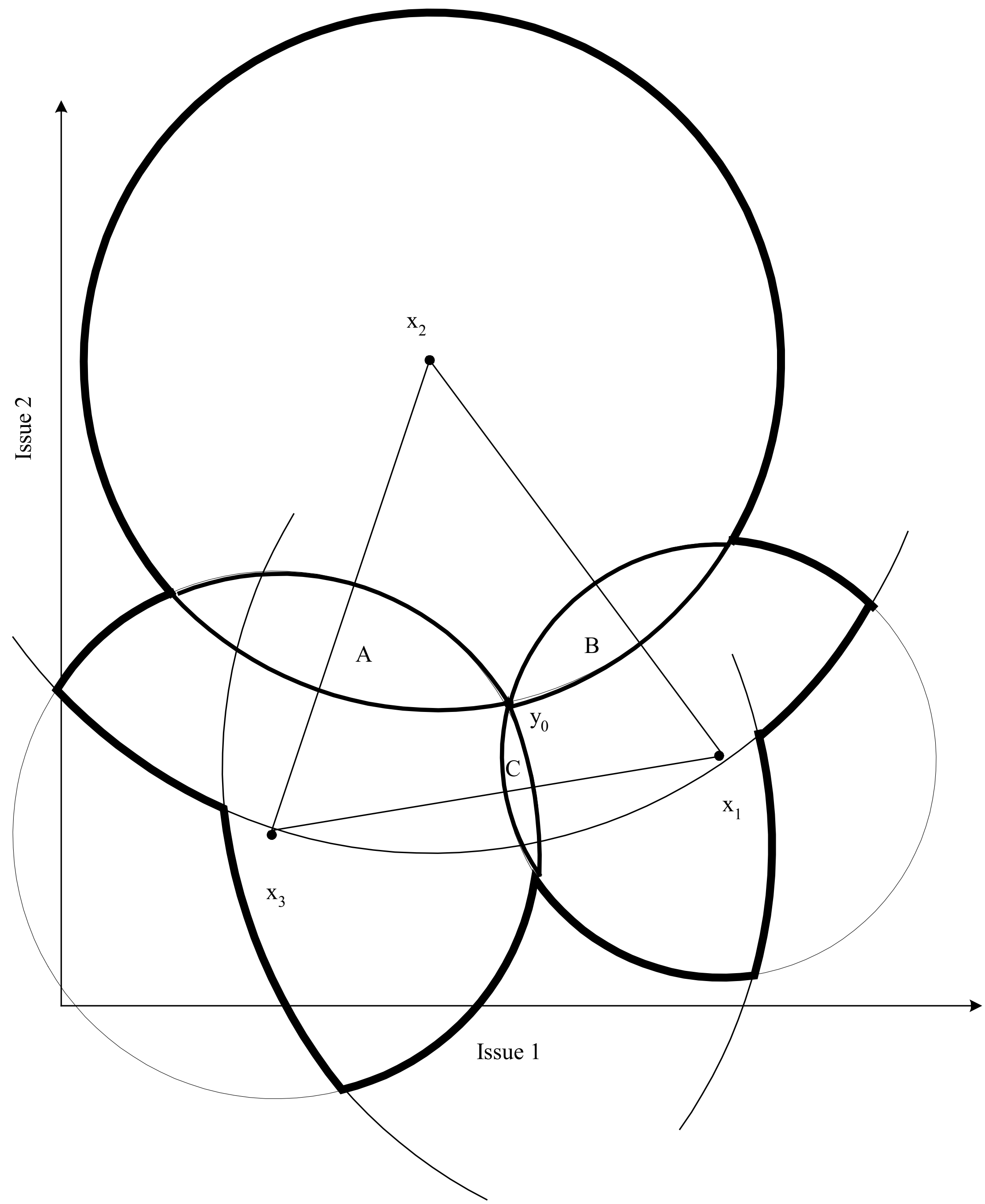

Figure 1: The Win Set and the Uncovered Set

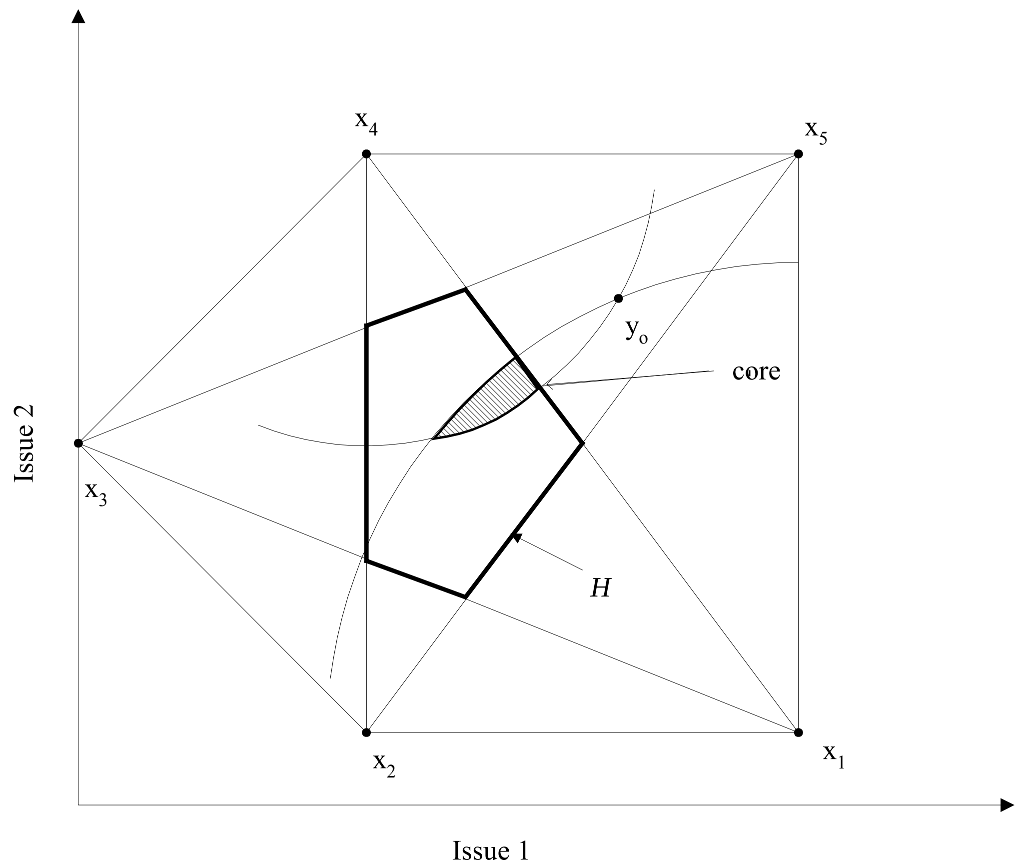

Figure 2: The core of a simple game with four-fifths rule

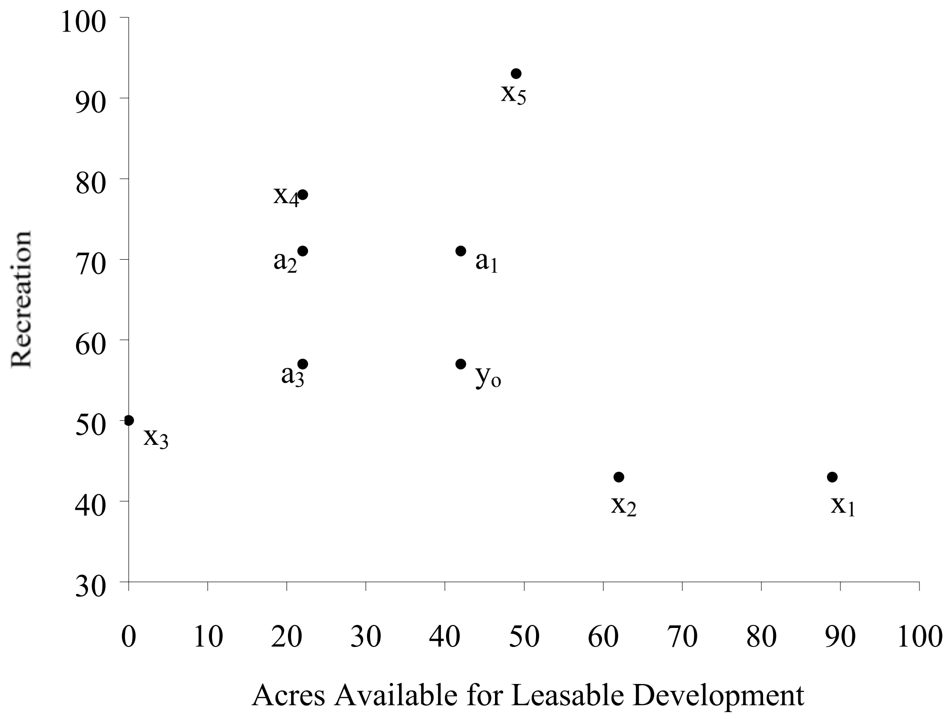

Figure 3: Stakeholders' ideal points

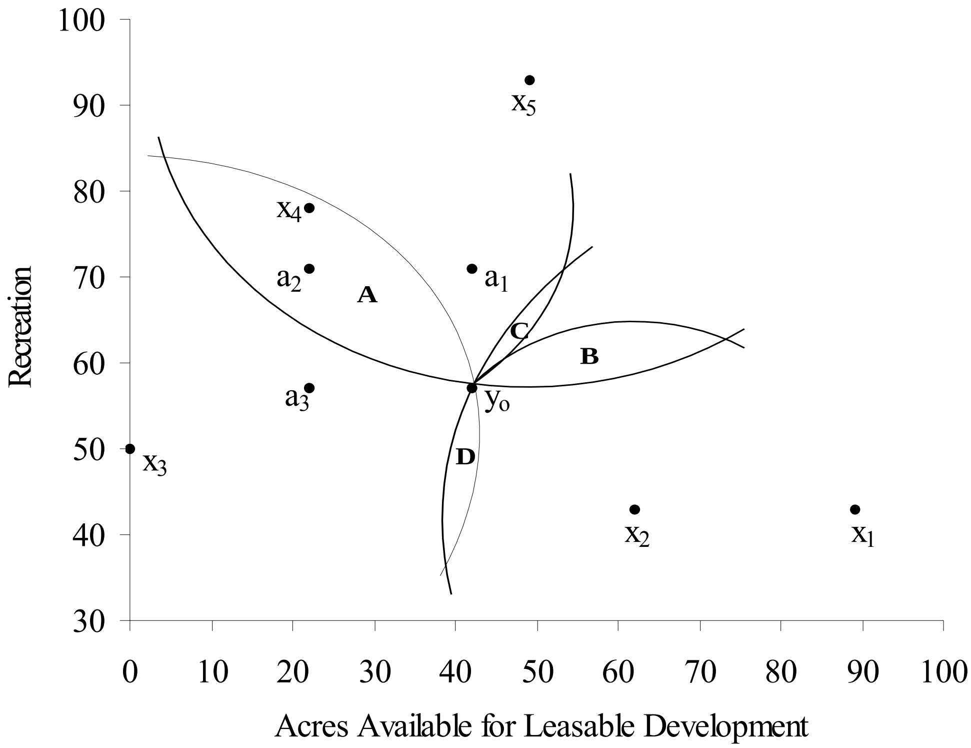

Figure 4: All win sets with a default alternative, y_o

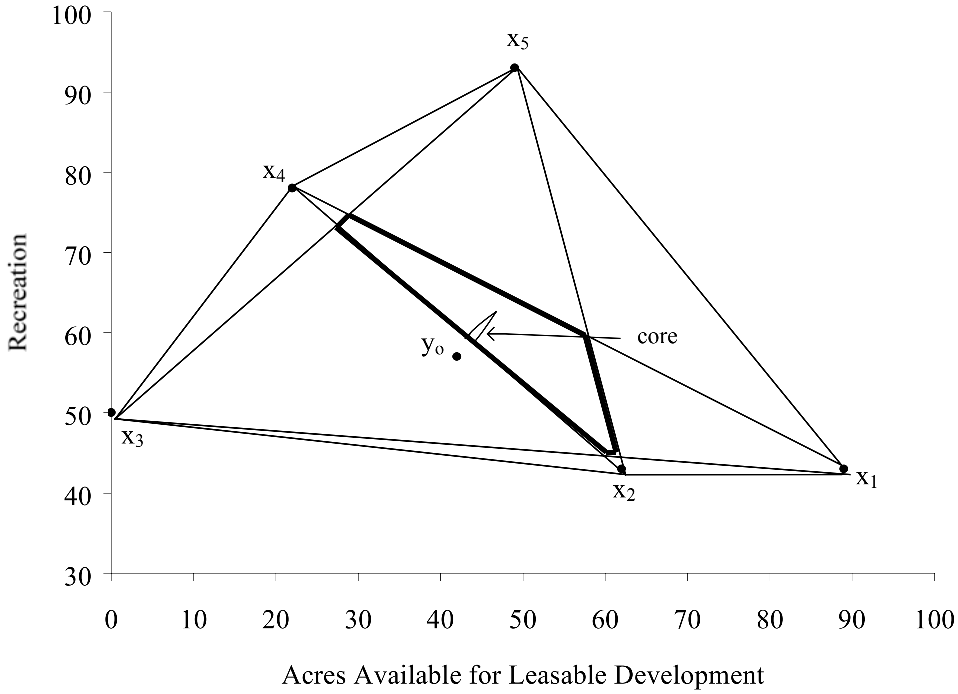

Figure 5: The core for a four-fifths supramajority rule game with a default alternative

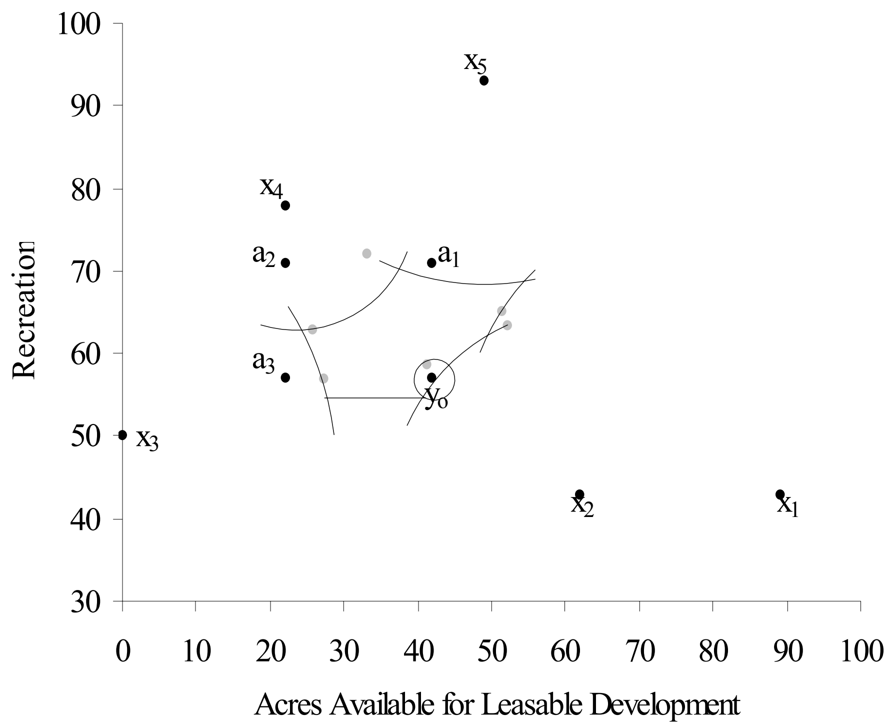

Figure 6: The Shapley Value for a seven stakeholder game

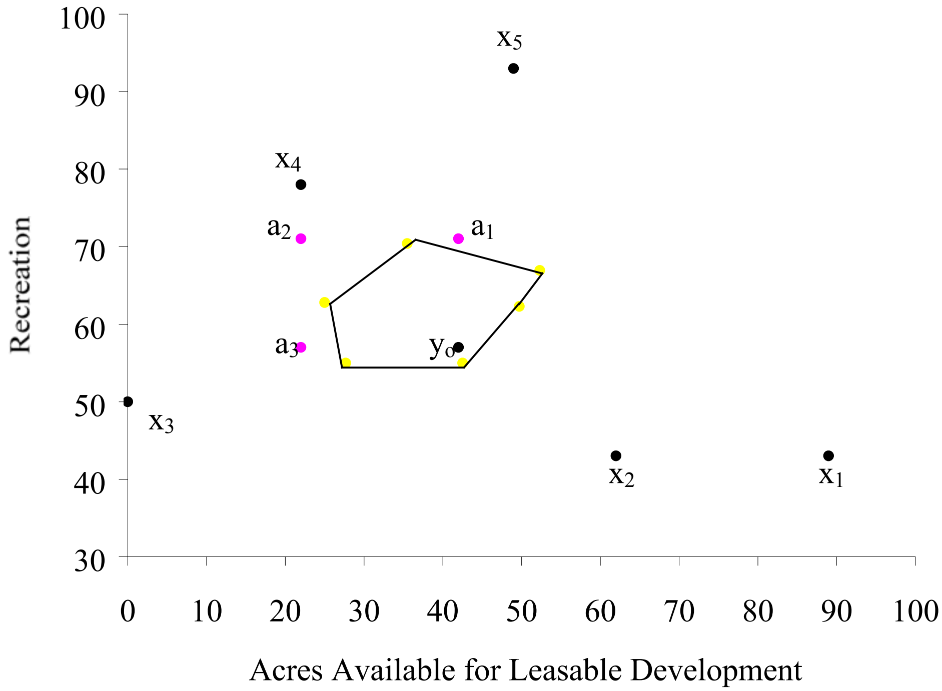

Figure 7: An approximate solution for the seven stakeholder Shapley Value

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